SNAPSMACK OPERABILITY AUDIT — DING DONG BELL

OPAUDIT 016 — A 16-bit file with an 8-bit picture inside it

Audit ID	2026-09-14-016
Report date	2026-09-14
First observed	2026-09-14, the first day a RAW file went through SNAP SLAPPER since the RAW handoff was built. Every edit in the month before was made from JPEGs.
Theme	SNAP SLAPPER is specified as a 16-bit non-destructive editor. It was 16-bit at the door and 8-bit in the room where the editing happens. The file it wrote said 16-bit; the picture inside had already been reduced to 8-bit.
Surfaces	SNAP SLAPPER's compositor (layers, blends, masks, effects); its TIFF export; the RAW handoff to RawTherapee; the "open developed file" step.
State	**FIXED / CLOSED** Corrected after the original report. The editor now reads high-bit sources into a float32 compositor and quantises once to genuine 16-bit output.
Watched live?	**Yes.** The original defect was reproduced on a real RAW. The rebuilt editor is now in regular use and working correctly; code and regression checks also prove 16-bit ingress, float32 editing, and real uint16 export.
Reporter	Sean (put the first real RAW through it and asked why the app still showed an 8-bit engine) + Codex (diagnosed it, once asked plainly, and did not dress it up).
Related	DING DONG BELL "Photo editor files opened in other editors" — the Photoshop difference test on that entry was run on an 8-bit source, so it proved the wrapper and not the bits; it stands, with that limit now written on it.
Publication	Public. No exploitable detail. SNAP SLAPPER has one user (Sean); no one else's files went through it.

1. SUMMARY

RawTherapee develops a RAW to a real 16-bit TIFF and hands it to SNAP SLAPPER correctly. SLAPPER's compositor is built on Pillow. Pillow has no multi-channel high-bit image mode, so the first thing the compositor does with a 16-bit image is convert it to 8-bit RGB. Every layer, blend, mask and effect then runs in 8-bit, and the result is written into a 16-bit TIFF. The container is honest; the contents are not. The original RAW was never touched — the editor is non-destructive at the source, and that held. But every edit of a high-bit image made through the app was quietly degraded. It went unnoticed for a month because the month's work was all JPEGs: 8-bit in, 8-bit composite, 8-bit out, nothing to lose, nothing visibly wrong.

2. WHAT A HUMAN SAW

Step	What happened
Open a RAW in SLAPPER	The app refused to develop it and offered only to open the original in RawTherapee. The RAW handoff (SLAPPER's sliders driving RawTherapee, a developed 16-bit TIFF coming back) had to be built first.
Handoff reported done — "6 RAW-focused tests passed"	The developed file would not open. The app tried to open the RawTherapee settings sidecar (`.pp3`) as if it were the photograph. Six tests had passed without one of them opening a developed image in the live app.
Ask why the app still shows an 8-bit engine	Codex diagnosed it: RawTherapee produces 16-bit, the Pillow compositor downgrades to 8-bit before any layer or effect runs. The "16-bit editor" claim was true at RAW decode and false at composite.

3. ROOT CAUSE — THREE OF THEM

The compositor never left 8-bit. Pillow can hold a single 16-bit channel or a float channel, but not 16-bit RGB. The whole editing stage therefore ran in 8-bit regardless of what it was handed. This is the defect that matters.

Files identified by guessing their names. The `.pp3`-opened-as-a-photo error, and a separate preview that snapped back to a stale cached TIFF, are the same bug: the pipeline works out which cached file is current by building a path string (add `.tif`, or open the `.pp3`) instead of tracking each file it makes by an assigned identity. Guess a filename and sooner or later you open the wrong file.

Done was claimed twice for a path nobody had run. Completion was reported — first for the handoff, then for RAW support — without a real RAW ever going through the live app. Green tests on the one workflow the feature exists for were taken as proof it worked. They were proof the tests ran.

4. EVIDENCE AT DIAGNOSIS

One real RAW, developed by RawTherapee to a 16-bit TIFF (confirmed at that step). SLAPPER's compositor entry point converting to Pillow `RGB` before layer work. Export writing a 16-bit TIFF from that 8-bit working image. The RAW test suite: six passing tests, none of which opened a developed file in the app. A month of exports, all from JPEG sources, all internally consistent and therefore silent.

5. WHAT WE DID

Resolution update, 2026-10-15: the compositor was rebuilt as specified and the editor is now working correctly in regular use. Its high-bit pipeline is:

Work in float32 internally; quantise to 16-bit only on export. Not 16-bit integer — that still clips highlights on the first curve. Float keeps the headroom that makes highlight recovery from a RAW possible at all.

Move colour compositing off Pillow. Pillow stays for 8-bit legacy reading and writing only.

Assigned identities for every file the pipeline makes. A typed registry where a settings sidecar can never be opened as an image and a preview proxy can never become an edit source. Same pass as the bit-depth work, because the two bugs share the cause.

6. VERIFIED — AND WHAT REMAINS

Closed 2026-10-15. A high-bit source is decoded into an H×W×C float32 array; layers, blends, masks, effects, geometry and adjustments remain in that 32-bit floating-point working space; highlight and shadow values are not clipped between operations. A 16-bit PNG or TIFF is quantised once, at the export boundary, to real unsigned 16-bit samples. The 8-bit image is now only the bounded screen preview and is never the authoritative editing source. Automated checks preserve 16-bit sample steps on ingress, assert float32 through the compositor and its filters, and inspect the exported file as uint16 with values reaching 65,535. Sean has also confirmed the rebuilt editor is working correctly in normal use.

7. WHAT WAS DAMAGED, AND WHAT WAS NOT

Safe	Original RAW files. Never modified.
Safe	The month of JPEG edits. 8-bit sources composited in 8-bit lost nothing they had to give.
Degraded	Any edit of a high-bit image made through the app: 8-bit data inside a 16-bit file. The RAW path — the path the tool exists for.

Caught on the first day of real RAW use rather than after a month of masters had gone through it.

8. EYEBALL NEXT

The separate Photoshop interchange claim still needs its own high-bit round-trip check. That is not a condition of this editor-engine repair: OPAUDIT 016 is closed because the editor's authoritative pipeline and finished 16-bit output are now proven. Other applications importing and returning those files remain separate interoperability claims and must be verified separately.

9. THE DOCTRINE THIS INCIDENT CREATED

> *A container is not its contents. A 16-bit file can hold an 8-bit picture. Checking that the output is a 16-bit TIFF proves the wrapper; prove the bits. And "done" on a feature is watching a real file go through the live app and come out right — a green test suite on a path nobody has walked certifies nothing.*